

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

MAY EXAMINATION

AMIS1012 ETHICS IN COMPUTING

THURSDAY, 7 MAY 2026

TIME: 2.00 PM – 4.00 PM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

DIPLOMA IN INFORMATION TECHNOLOGY

DIPLOMA IN INFORMATION SYSTEMS

DIPLOMA IN SOFTWARE ENGINEERING

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AMIS1012 ETHICS IN COMPUTING**Question 1**

- a) Explain **THREE (3)** common ethical issues faced by IT users. (9 marks)
- b) Describe **morals, ethics, and laws** in computing. (6 marks)
- c) Describe **passive attacks** and **active attacks**. (4 marks)
- d) Explain **TWO (2)** types of cybercrime. (6 marks)

[Total: 25 marks]

Question 2

A large e-commerce company stores customer records, transaction data, and internal business systems on its IT infrastructure. Recently, management has been concerned about cyber threats and wants to strengthen risk management and cybersecurity practices.

Using the above scenario, answer the following questions:

- a) Describe the first two steps of the security risk assessment process for the company. (6 marks)
- b) Analyse **THREE (3)** ways the CIA Triad can be applied to safeguard the company's information systems. (9 marks)
- c) Explain **TWO (2)** pillars of trustworthy computing and their relevance to the company. (6 marks)
- d) Explain how the company can use ISO 27001 and the NIST Cybersecurity Framework to improve its cybersecurity practices. (4 marks)

[Total: 25 marks]

Question 3

A software company is developing an online banking application for financial services. The system must be secure, reliable, and error-free before its release. The company aims to improve its development process, testing strategy, and overall software quality.

Using the above scenario, answer the following questions:

- a) Recommend a suitable software development methodology. Justify your choice. (4 marks)
- b) Analyse **THREE (3)** effects of failing to detect software defects early in the banking application. (9 marks)
- c) Explain how the company can use manual testing and automation testing to test the banking system. (6 marks)

AMIS1012 ETHICS IN COMPUTING**Question 3 (Continued)**

- d) Explain how **THREE (3)** software quality attributes help improve the quality of the banking application. (6 marks)

[Total: 25 marks]

Question 4

A group of young developers launches an indie online game that becomes viral on social media. The game includes original characters, music, logos, and unique gameplay features. As the game becomes popular, other companies start copying its design and branding.

Using the above scenario, answer the following questions:

- a) Explain **TWO (2)** reasons why Intellectual Property Rights are important for the game creators. (6 marks)
- b) Explain **TWO (2)** intellectual property issues that may occur if the developers fail to protect their game. (6 marks)
- c) Describe **THREE (3)** types of Intellectual Property Rights that the developers can use to protect their game. (6 marks)
- d) Briefly explain **TWO (2)** actions the developers should take when their intellectual property is infringed. (4 marks)
- e) Explain why downloading pirated digital games is considered an intellectual property violation. (3 marks)

[Total: 25 marks]